HARMILAP SINGH DHALIWAL

AI Engineering · Agentic AI Platforms · LLMOps in Regulated Finance

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AI & Data Science Lead in RBC's CFO Group, architecting and building production AI and LLM systems end-to-end. In the last year I shipped RBC's first bank-wide production AI agent, brought a multi-stage LLM analytics platform to production, and shipped a senior-leadership analytics prototype over a single weekend. The engineering surface spans agentic state graphs, multi-stage LLM pipelines with defense-in-depth gating, multi-layer RAG, foundation-model-agnostic architecture, and the LLMOps practice that holds the production bar across all of it. Eight years of progressive AI/ML engineering — three-plus of them shipping production AI in regulated finance, the most recent eighteen months on agentic LLM systems. Increasingly focused on what it takes for these patterns to scale beyond a single business line, into platforms the rest of the bank can build on.

PROFESSIONAL EXPERIENCE

AI & Data Science Lead, Finance — Royal Bank of Canada *April 2025 – Present*

CFO Group, Enterprise Finance · Toronto, ON

Manage 1 Senior AI Scientist direct and 2 interns onboarded May 2026 (9 cumulative, peak 5 simultaneous). Contribute to AI/ML hiring decisions across CFO Group since 2023 (university recruiting, candidate screening, performance reviews). Lead cross-functional production-deployment partnership with Global Functions Technology (GFT) on OpenShift via CI/CD. Hands-on across ~70% of all project work.

AI Platforms & Production Systems

- Conceived, architected, and shipped RBC's first bank-wide production AI agent (PAR ASSIST). Pilot launched April 2026; deployed bank-wide May 2026. The system guides project authors through Project Approval Requests across every business line, compressing approver cycle time, reducing submitter-approver back-and-forth, and surfacing overage risk at drafting before proposed costs exceed budget thresholds.
- Owned the full engineering stack of PAR Assist: ETL, agentic state graph (LangGraph), MCP tool layer (template selection, field assignment, conflict resolution, ambiguity detection), PostgreSQL/pgvector with custom multi-layer RAG over uploaded documents and institutional knowledge, and frontend.
- Unlocked self-service financial analytics for CFO Group leadership and finance teams bank-wide through ASTRAEUS. Production since November 2025. Brought headcount-movement queries from days (line-of-business level only) to real time at any granularity across ~40K cost centres and ~9K rollups, eliminating the wait-for-custom-report bottleneck across HR cost, headcount, and open-positions domains.
- Architected ASTRAEUS as a multi-stage LLM pipeline (gate → parallel domain metadata extraction across EPM, Headcount, Open Positions → answer or cross-domain synthesis) with deterministic Python orchestration and defense-in-depth gating between stages (relevance, injection, scope, privilege). Foundation-model-agnostic by design; migrated cleanly from Command-A to GPT-4.1 once bank-wide approval landed for sensitive data.
- Enforced fine-grained, least-privilege data rights across the dashboard, chatbot, and report surfaces, with audit trails carried through every stage.
- Collapsed senior-leadership query turnaround from minutes-to-hours to seconds by shipping ARGUS, an analytics agent prototype generating on-demand visuals and dashboards. Solo weekend build inside RBC Assist Pro's fixed orchestration layer; GPT-5 vision plus Claude Opus 4.6.
- Built the Aegis v2 benchmarking module solo in 2 weeks while running Astraeus and the 2025 Amplify intern program in parallel: multi-stage RAG with multi-gate parsing across bank, parameter, platform, and time-period dimensions over Big 6 Canadian bank Supplementary Financial Package data, plus the metadata and embedding layer foundation. Productionalized by direct report.

Platform Practice, Engineering Leadership & Mentorship

- Own the LLMOps surface across CFO Group AI systems: multi-stage pipeline orchestration, inter-stage validation contracts, defense-in-depth gating, regression testing on foundation-model upgrades, and observability across multi-stage LLM systems.

- Carry production AI systems through Model Risk Management and responsible-AI review with end-to-end accountability across architecture, evaluation evidence, monitoring, and post-deployment performance.
- Fold each significant advance in the agentic AI stack (MCP, multi-stage LLM pipelines with stage-level gating, multi-layer RAG, current-generation foundation models including Claude Opus 4.6, GPT-5 vision, GPT-4.1) into shipped CFO Group systems within weeks of stable availability.
- Mentor Senior AI Scientists and interns through architecture reviews on agentic system design, RAG configuration, and SQL guardrails. Instituted GitHub-first engineering practices (PRs, code reviews, refactoring standards) that enabled junior contributors to ship production code.
- Lead the 2025 Amplify intern program (4 interns) and the 2026 cohort (2 interns onboarded May 2026): scoped problems including the PAR Assist concept I subsequently built end-to-end, reviewed code, coached executive presentations.

Enterprise Finance Senior Data Scientist — Royal Bank of Canada *September 2022 – April 2025*

CFO Group, Enterprise Finance · Toronto, ON

- Reduced commodity tax return preparation from months to 90 minutes across ~$250M GST and ~$350M PVAT by architecting the Commodity Tax automation pipeline (PySpark on CDP over enterprise GL Journal data). CFO Group RBC Quarterly Team Award (Q4 2023).
- Unblocked timely Big 6 peer benchmarking for CFO Group leadership by building Aegis v1 end-to-end. Critical unlock: solo-built automated extraction and matching logic for Supplementary Financial Packages whose schemas shift quarterly, the long-standing bottleneck that had blocked timely peer analysis. CFO One RBC Team Award (2025).
- Built EDS Automation, a PAR-actual-vs-planned financial comparison system across financial accounts, benefits, expenses, capital expenditure, revenue, and depreciation. Custom Python pipeline scheduled in Dataiku, Tableau dashboards for business consumption. Widely adopted across the finance team.
- Engineered manual journal entry automation on PySpark + CDP; built the EUDA automation dashboard, CFO Group's single source of truth for End-User Computing Application status.

Machine Learning Engineer — Quantiphi Inc. *October 2021 – September 2022*

Toronto, ON

- Lifted document-understanding accuracy from ~70% to 99.95% by productionizing the Humana pipeline on GCP: Google Document AI OCR, OpenCV pixel-level checkbox detection, Random Forest classification, BigTable + BigQuery infrastructure.
- Integrated custom and AutoML models on Vertex AI into Humana's production review workflows. Earned Google Associate Cloud Engineer during this tenure.

Data Scientist — Tata Consultancy Services *August 2016 – November 2019*

Pune, India

- Delivered $3M annual cost savings and measurable NOx/SOx/CO emissions reduction at Mitsubishi Hitachi Power Systems' Maizuru 900MW coal plant (Japan) by architecting the Combustion Tuning Digital Twin end-to-end as sole data scientist on a 3-person R&D team: ETL across 90+ sensors, 84 regression models, Particle Swarm Optimization producing closed-loop control recommendations.
- Prevented brittle models from reaching the optimization layer by designing the multi-indicator evaluation framework (k-fold cross-validation, R^2, RMSE, MAPE, fold variance) qualifying each of the 84 production models.
- Shipped supporting production models: LSTM time-series forecasting for Ammonium Bisulphate deposition (maintenance planning), XGBoost Transformer Life Prediction, and a 5-algorithm coal classification ensemble.
- Won 2nd place out of 600 in TCS's CV Hackathon for a deep-learning math notation detector (Innovation Pride Award, September 2019).

TECHNICAL SKILLS

AI Platform & Agentic AI	Agentic state graphs (LangGraph, LangChain), MCP (Model Context Protocol), intent-routing patterns, MCP tool design, agentic state management, foundation-model-agnostic architecture
LLMOps	Multi-stage LLM pipeline orchestration, inter-stage validation contracts, defense-in-depth gating, regression testing on foundation-model upgrades, multi-stage LLM observability, multi-layer RAG, PostgreSQL/pgvector, prompt engineering at production scale, evaluation pipelines for regulated production
LLM Engineering	Production deployment (Claude Opus 4.6, GPT-4.1, GPT-5 vision, Command-A), text-to-SQL, intent parsing, KPI disambiguation, LLM evaluation, guardrailed generation
Responsible AI & Governance	Model Risk Management (MRM) review, multi-level privilege checks, audit trails across multi-stage pipelines, entitlement modeling (EPM-to-SQL translation), regulated-environment quality bar
Traditional ML & Deep Learning	PyTorch, TensorFlow, scikit-learn, XGBoost, supervised/unsupervised learning, time series (LSTM, RNN), CNNs, Transformers, NLP (BERT), Particle Swarm Optimization, k-fold validation
Programming & Data Engineering	Python, R, SQL, PySpark on Cloudera Data Platform (CDP), ETL pipeline design
Cloud & Platforms	Google Cloud Platform, Vertex AI, AutoML, BigQuery, BigTable, OpenShift (via GFT CI/CD), Dataiku, Tableau

EDUCATION & CERTIFICATIONS

Post Graduate Professional Certificate, Big Data Analytics *January 2021 – August 2021*
Georgian College, Barrie, Ontario

Bachelor of Engineering, Electronics & Communications Engineering *July 2012 – June 2016*
Thapar University, Patiala, India

CERTIFICATIONS DeepLearning.AI Deep Learning Specialization · TensorFlow Developer Specialization · Johns Hopkins Data Science Specialization

AWARDS & RECOGNITION

- **CFO One RBC Team Award (2025)** — LLM/AI initiatives, Aegis v1 productionization for Big 6 Canadian bank benchmarking.
- **CFO Group RBC Quarterly Team Award (Q4 2023)** — Commodity Tax automation, $600M+ allocation, months to 90 minutes.
- **Innovation Pride Award, TCS (September 2019)** — 2nd of 600 in TCS CV Hackathon, deep learning math notation detector.